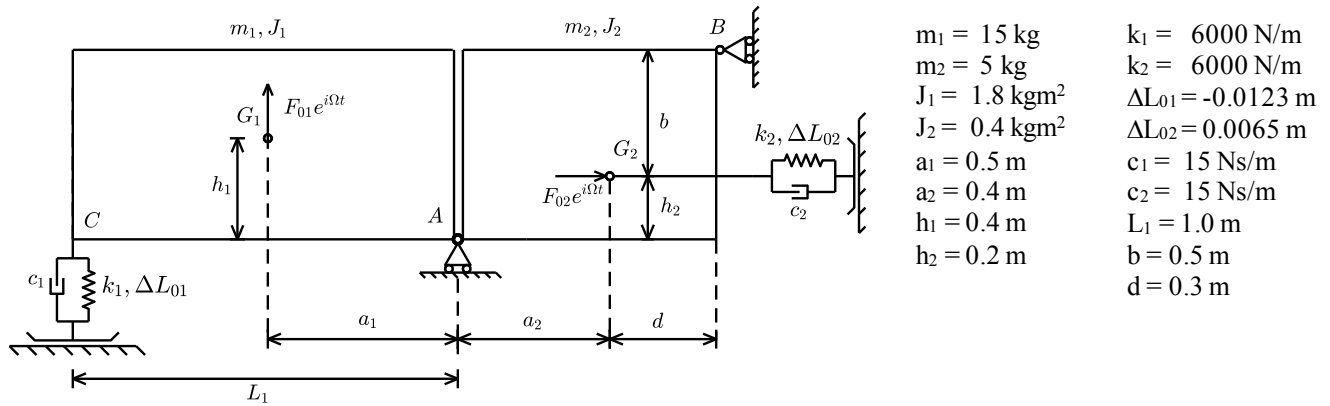


Problem 1:



The mechanical system above lies in the vertical plane, is shown in its equilibrium position. Both rigid bodies are connected to the cart A and can rotate without touching each other.

The students are requested to:

0. Write on top of each page their surname, name, Person Code and **SNxxxx code** (see below)
1. Write the equations of motion of the system linearized around the given equilibrium configuration (provide this answer on paper: Jacobian matrices and second-order stiffness terms, together with at least the main kinematic relationships that justify the solution)
2. Compute the natural frequencies and modal shapes (provide this answer on paper, copying the values from MATLAB);
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic vertical force $F_{01}e^{i\Omega t}$. Output: the vertical component of the displacement of the centers of mass G_1 and G_2 . Save the graphical results as SNxxxx1.fig and SNxxxx2.fig, with SNxxxx your identifier, see below;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic horizontal force $F_{02}e^{i\Omega t}$. Output: the forces in the carts A and B. Save the graphical result as SNxxxx3.fig and SNxxxx4.fig.
5. Compute the static preloads ΔL_{01} and ΔL_{02} when the mass m_2 is 10 kg (provide the answer on paper).

The answer to points 1, 2 and 5 shall be provided on paper (for point 2 please copy on paper the values of the natural frequencies and eigenmodes produced by the MATLAB script); the answer to points 3 – 4 shall be provided by delivering the MATLAB script and .fig files as a compressed archive uploaded on WeBeeP, see instructions below.

You must upload one single archive file for both problem 1 and problem 2, which should contain your MATLAB scripts and all the .fig files you want to deliver as part of your solution.

- i.) Accepted formats for the archive files are: .7z .rar .zip;
- ii.) The file must be named SNxxxx with “S” the first letter in your surname, “N” the first letter in your name (in case of multiple surnames/names just consider the first one), “xxxx” the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the name of the file should be: BS5678.ext;
- iii.) The file must be uploaded as an assignment on WeBeeP, the name of the assignment is **Test_11_Jan_2022**;
- iv.) the deadline to upload the assignment will be announced during the written exam.

Your identifier: SNxxxx with S the first letter of your surname, N the first letter of your name and xxxx the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the identifier BS5678

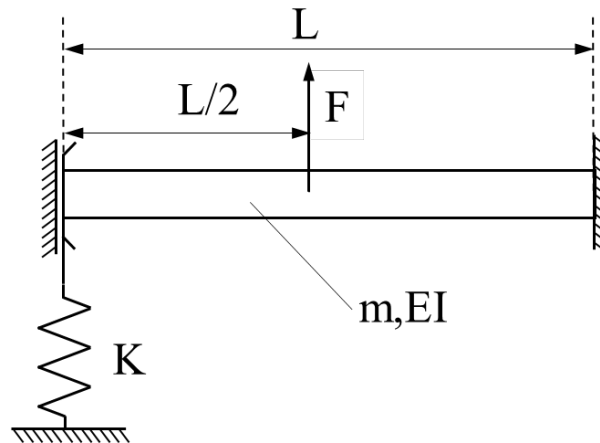
Please be aware that your answers to the test will not be considered in case:

- a. the information stated at point 0 above is missing;
- b. your archive file is uploaded in the wrong folder
- c. the name of the archive file is different from the identifier SNxxxx described above;

For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one.

Problem 2:

Consider the bending motion of a beam with uniform section and material properties having length L . The left end of the beam is constrained by a slider and the right end is constrained by a clamp. A lumped spring with stiffness K is attached to the left end of the beam.



DATA:

$$EI = 4 \times 10^5 \text{ Nm}^2$$

$$m = 10 \text{ kg/m}$$

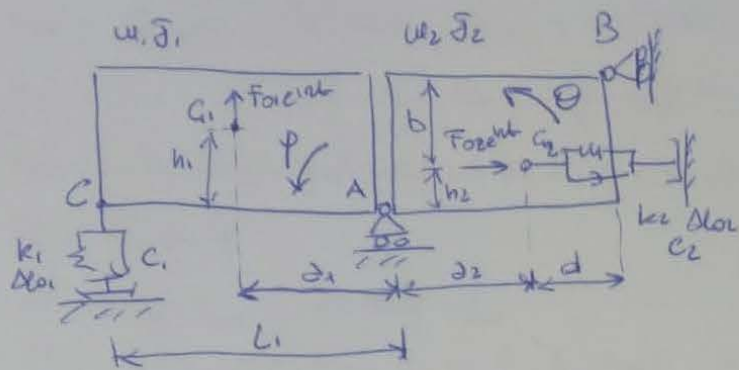
$$L = 4 \text{ m}$$

$$K = 1 \times 10^6 \text{ N/m}$$

- 1) Write the boundary conditions and the expression of the square matrix of coefficients H of the linear homogeneous system defining the natural frequencies of the system. Deliver this part of the solution on paper.
- 2) Compute the natural frequencies of the system in the frequency range 0-200 Hz and the related modal shapes. Deliver the values of the natural frequencies on paper and include in your archive file one .fig file for each modal shape. The .fig files shall be named SNxxxxP2i with SNxxxx your identifier (see below) and $i=1,2,3$ (use as many as needed);
- 3) Plot the Bode diagram (log y scale) of the magnitude of the FRF for
 - input: force having unit magnitude applied in the centre of beam
 - output: vertical vibration in the centre of beam (co-located FRF).

The Bode diagram shall be delivered as a .fig file named SNxxxxP2FRF included in your archive.

PROBLEM 1

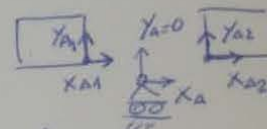


$$\underline{x} = \begin{Bmatrix} \theta \\ \varphi \end{Bmatrix} ; \underline{x}_0 = \begin{Bmatrix} \phi \\ \rho \end{Bmatrix}$$

• 2 rigid bodies : 6 dof.

• 1 cart B : - 1 dof.

• cart A :



$$\begin{cases} y_{A1} = y_{A2} = 0 \\ y_{A1} = y_{A2} = 0 \\ x_{A1} = x_{A2} = x_A \end{cases} \rightarrow 3 \text{ dof.}$$

2 dof.

$$T = \frac{1}{2} m_1 \dot{x}_{G1}^2 + \frac{1}{2} m_2 \dot{x}_{G2}^2 + \frac{1}{2} I_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{G2}^2 + \frac{1}{2} I_2 \dot{\varphi}^2 + \frac{1}{2} \dot{\theta}^2$$

$$D = \frac{1}{2} \sum_{i=1}^2 c_i \Delta l_i^2 ; V_{el} = \frac{1}{2} \sum_{i=1}^2 k_i \Delta l_i^2 ; V_g = \sum_{i=1}^2 m_i g h_{Gi}$$

$$\delta W = \vec{F}_{01} e^{int} \times \delta \vec{x}_{G1} + \vec{F}_{02} e^{int} \times \delta \vec{x}_{G2} = \vec{F}_{01} e^{int} \delta x_{G1} + \vec{F}_{02} e^{int} \delta x_{G2}$$

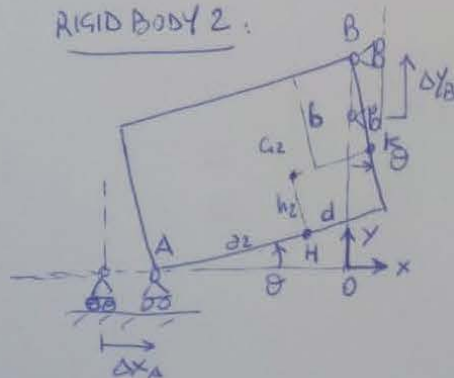
$$[w_{PR}] = \text{diag}(m_1, m_1, I_1, m_2, m_2, I_2) ; \underline{y}_M = \{x_{G1}, x_{G1}, \theta, x_{G2}, x_{G2}, \varphi\}^T$$

$$[c_{PR}] = \text{diag}(c_1, c_2) ; [k_{PR}] = \text{diag}(k_1, k_2) ; \underline{y}_0 = \underline{y}_k = \{\Delta l_1, \Delta l_2\}^T$$

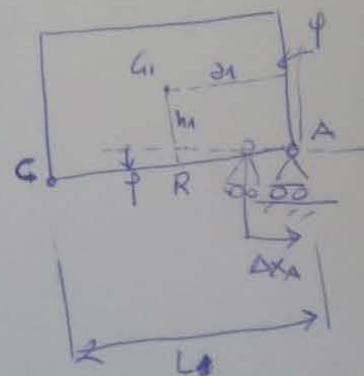
$$\underline{F} = \{F_{01}, F_{02}\}^T e^{int} ; \underline{y}_d = \{x_{G1}, x_{G2}\}^T$$

Non-linear kinematic relationships :

RIGID BODY 2 :



RIGID BODY 1 :



$$X_{G2} = \bar{B}k \sin \theta - \bar{G}_2 k \cos \theta = b \sin \theta - d \cos \theta$$

$$Y_{G2} = \bar{A}h \sin \theta + h_2 \cos \theta = a_2 \sin \theta + h_2 \cos \theta$$

$$X_A = (\bar{B}k + \bar{G}_2 h) \sin \theta - (\bar{G}_2 k + \bar{A}h) \cos \theta = (b + h_2) \sin \theta - (d + a_2) \cos \theta$$

$$X_{G1} = X_A - \bar{A}R \cos \phi - \bar{G}_1 R \sin \phi = (b + h_2) \sin \theta - (d + a_2) \cos \theta - a_1 \cos \phi - h_1 \sin \phi$$

$$Y_{G1} = -\bar{A}R \sin \phi + \bar{G}_1 R \cos \phi = -a_1 \sin \phi + h_1 \cos \phi$$

$$Y_C = -L \sin \phi$$

$$\Delta L_1 = Y_C - Y_{G1} = -L \sin \phi - Y_{G1}$$

$$\Delta L_2 = -(X_{G2} - X_{G1}) = -a_2 \sin \theta - h_2 \cos \theta - X_{G2}$$

$$\frac{\partial X_{G2}}{\partial \theta} = b \cos \theta + d \sin \theta \quad ; \quad \frac{\partial X_{G2}}{\partial \phi} = 0$$

$$\frac{\partial Y_{G2}}{\partial \theta} = a_2 \cos \theta - h_2 \sin \theta \quad ; \quad \frac{\partial Y_{G2}}{\partial \phi} = 0$$

$$\frac{\partial X_{G1}}{\partial \theta} = (b + h_2) \cos \theta + (d + a_2) \sin \theta \quad ; \quad \frac{\partial X_{G1}}{\partial \phi} = a_1 \sin \phi - h_1 \cos \phi$$

$$\frac{\partial Y_{G1}}{\partial \theta} = 0 \quad ; \quad \frac{\partial Y_{G1}}{\partial \phi} = -a_1 \cos \phi - h_1 \sin \phi$$

$$\frac{\partial \Delta L_1}{\partial \theta} = 0 \quad ; \quad \frac{\partial \Delta L_1}{\partial \phi} = -L \cos \phi$$

$$\frac{\partial \Delta L_2}{\partial \theta} = -\frac{\partial X_{G2}}{\partial \theta} \quad ; \quad \frac{\partial \Delta L_2}{\partial \phi} = -\frac{\partial X_{G2}}{\partial \phi}$$

$$[\Lambda_{cu}] = \begin{bmatrix} b + h_2 & -h_1 \\ 0 & -a_1 \\ 0 & 1 \\ b & 0 \\ a_2 & 0 \\ 1 & 0 \end{bmatrix} \quad [\Lambda_u] = [\Lambda_c] = \begin{bmatrix} 0 & -L \\ -b & 0 \end{bmatrix}$$

$$[\Lambda_e] = \begin{bmatrix} 0 & -a_1 \\ b & 0 \end{bmatrix}$$

$$\begin{cases} h_{s1} = l_{s1} \\ h_{s2} = l_{s2} \end{cases}$$

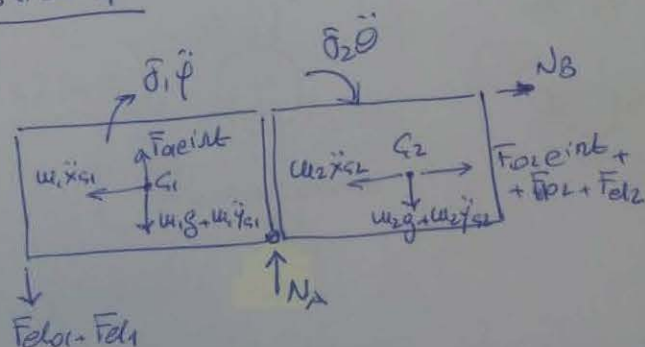
$$\begin{cases} \frac{\partial^2 h_{s1}}{\partial \theta^2} = \frac{\partial^2 h_{s1}}{\partial \theta \partial \varphi} = 0 \\ \frac{\partial^2 h_{s1}}{\partial \varphi^2} = c_1 \sin \varphi - h_1 \cos \varphi \end{cases} \rightarrow [k_{s1}] = m_1 g \begin{bmatrix} 0 & 0 \\ 0 & -h_1 \end{bmatrix}$$

$$\begin{cases} \frac{\partial^2 h_{s2}}{\partial \theta^2} = -c_2 \sin \vartheta - h_2 \cos \vartheta \\ \frac{\partial^2 h_{s2}}{\partial \varphi^2} = \frac{\partial^2 h_{s2}}{\partial \theta \partial \varphi} = 0 \end{cases} \rightarrow [k_{s2}] = m_2 g \begin{bmatrix} -h_2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{cases} \frac{\partial^2 \Delta L_1}{\partial \theta^2} = \frac{\partial^2 \Delta L_1}{\partial \theta \partial \varphi} = 0 \\ \frac{\partial^2 \Delta L_1}{\partial \varphi^2} = L \sin \varphi \end{cases} \rightarrow [k_{el II, 1}] = [0]$$

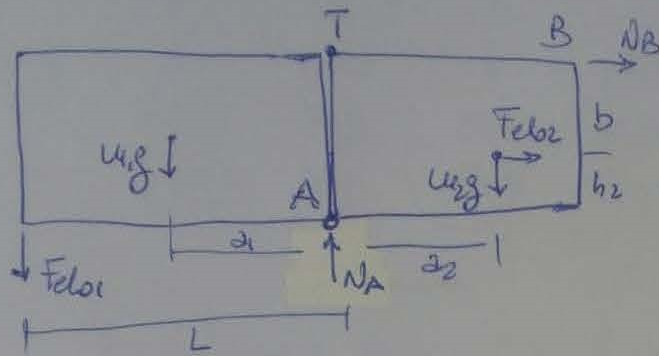
$$\begin{cases} \frac{\partial^2 \Delta L_2}{\partial \theta^2} = b \sin \vartheta - d \cos \vartheta \\ \frac{\partial^2 \Delta L_2}{\partial \varphi^2} = \frac{\partial^2 \Delta L_2}{\partial \theta \partial \varphi} = 0 \end{cases} \rightarrow [k_{el II, 2}] = k_2 \Delta L_2 \begin{bmatrix} -d & 0 \\ 0 & 0 \end{bmatrix}$$

• Question 4:



$$\sum F_x = 0: N_B = m_1 \ddot{x}_{s1} + m_2 \ddot{x}_{s2} - \underbrace{F_{el02} - F_{el2} - F_{el_eint}}_{\text{static}}$$

$$\sum F_y = 0: N_A = \underbrace{(m_1 g + m_2 g + F_{el01})}_{\text{static}} + m_1 \ddot{y}_{s1} + m_2 \ddot{y}_{s2} + F_{el1} - F_{el_eint}$$



$$\sum M_A^{\text{BODY 1}} = 0 :$$

$$F_{o1} L + w_1 g d_1 = 0$$

$$\rightarrow \Delta_{o1} = - \frac{w_1 g}{k_1} \frac{d_1}{L_1}$$

$$\left\{ \begin{array}{l} \sum F_x = 0 \rightarrow N_B + F_{elo2} = 0 \\ \sum M_A^{\text{BODY 2}} = 0 \rightarrow N_B(b+h_2) + F_{elo2} h_2 + w_2 g d_2 = 0 \end{array} \right.$$

$$\begin{bmatrix} b+h_2 & h_2 \\ 1 & 1 \end{bmatrix} \begin{Bmatrix} N_B \\ F_{elo2} \end{Bmatrix} = \begin{Bmatrix} -w_2 g d_2 \\ 0 \end{Bmatrix} \rightarrow \Delta_{o2} = \frac{F_{elo2}}{k_2}$$

$$\text{or: } \sum M_T^{\text{BODY 2}} = 0 : F_{elo2} b - w_2 g d_2 = 0 \rightarrow \Delta_{o2} = \frac{w_2 g}{k_2} \frac{d_2}{b}$$